1 Orthogonal Projection

Definition 1.1 (Orthogonal projection)**.** Suppose U is a finite-dimensional subspace of V. The *orthogonal projection* of V onto U is the operator $P_U \in \mathscr{L}(V)$ defined as follows: For each $v \in V$, write $v = u + w$, where $u \in U$ and $w \in U^\perp$. Then let $P_U v = u$.

Exercise 1.2. Explain why P_U is well-defined.

Example 1.3 (Orthogonal projection onto a one-dimensional subspace)**.** Suppose $u \in V$ with $n \neq 0$ and U is the one-dimensional subspace of V defined by $U = \text{span}\{u\}$.

If $v \in V$, then

$$v = \frac{(v,u)}{||u||^2}u + \left(v - \frac{(v,u)}{||u||^2}u\right),$$

where the first term on the right is in $\text{span}\{u\}$ (and thus is in U) and the second term on the right is orthogonal to u (and thus is in $U^\perp$). Thus $P_U v$ equals the first term on the right. In other words, we have the formula

$$P_U v = \frac{(v,u)}{||u||^2}u,$$

for every $v \in V$.

The formula above becomes $P_U u = u$ if $v = u$ and becomes $P_U v = 0$ if $v \in \{u\}^\perp$. These equations are special cases of (b) and (c) in the next result.

Proposition 1.4 (Properties of orthogonal projection P_U)**.** *Suppose U is a finite-dimensional subspace of V. Then*

(a) $P_U = \mathscr{L}(V)$

(b) $P_U u = u$ *for every* $u \in U$

(c) $P_U w = 0$ *for every* $w \in U^\perp$

(d) $\text{im}\, P_U = U$

(e) $\ker P_U = U^\perp$

(f) $v - P_U v \in U^\perp$ *for every* $v \in V$

(g) $P_U^2 = P_U$

(h) $||P_U v|| \leq ||v||$ *for every* $v \in V$

(i) If $\{e_1, \ldots, e_m\}$ *is an orthonormal basis of* U *and* $v \in V$*, then*

$$P_U v = \sum_{i=1}^{n} (v, e_i) e_i.$$

Exercise 1.5. Prove Proposition 1.4.

2 Minimization Problems

The following problem arises frequently: Given a subspace U of V and a point $v \in V$, find a point $u \in U$ such that $||v-u||$ is as small as positive. The next result shows that $u = P_U v$ is the unique solution of this minimization problem.

Proposition 2.1. *Suppose U is a finite-dimensional subspace of V, $v \in V$, and $u \in U$. Then*

$$||v - P_U v|| \leq ||v - u||.$$

Furthermore, the inequality above is an equality if and only if $u = P_U v$.

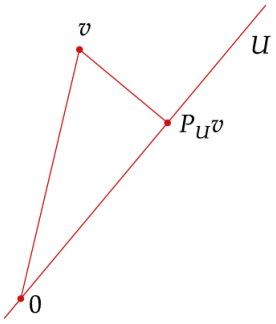


Figure 1: $P_U v$ is the closest point in U to v.

Exercise 2.2. Prove Proposition 2.1.

Proposition 2.1 is often combined with Proposition 1.4(i) to compute explicit solutions to minimization problems, as in the following exercise.

Exercise 2.3. Use Proposition 2.1 to find a polynomial u with real coefficients of degree at most 5 that approximates the sine function as closely as possible on the interval $[-\pi,\pi]$, in the sense that

$$\int_{-\pi}^{\pi} |\sin(x) - u(x)|^2 dx$$

is minimized. Compare your answer to $\sin(x)$ as well as the Taylor polynomial $p(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!}$ on $[-\pi,\pi]$ by using Mathematica or Desmos to plot $\sin(x), u(x)$, and $p(x)$ on the interval $[-\pi,\pi]$. (Note: I do not recommend doing this problem by hand as it is very tedious. A solution is provided in a Mathematica notebook on Canvas for you to check your work.)